

FTH Cosmology: Unifying Finsler Geometry with Timescape Inhomogeneities for Dark Energy

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Abstract

The standard Λ CDM model attributes late-time cosmic acceleration to a cosmological constant Λ whose observed magnitude lacks a first-principles derivation and exhibits persistent tensions with late-universe measurements of H_0 and high-redshift observations from JWST. We present the Finsler–Timescape Hybrid (FTH) framework, a mathematically closed extension of inhomogeneous cosmology that replaces Λ as a free vacuum-energy parameter with a direction-dependent kinematic backreaction Q_D^F derived from a single variational principle on the tangent bundle TM .

The FTH action is restricted to the flat void sector V_{flat} , characterized by trivial fiber topology $e(TM)=0$, spatial flatness $k_D=0$, and scale separation $r_V \ll r_{\text{inj}}$. The Randers dipole amplitude $b_0(z)$ evolves via a torsion-driven Riccati flow anchored exclusively to the Planck 2018 CMB kinematic dipole, yielding $b_0(z=14) \approx 0.030$ with no freely tunable parameters.

Fiber integration over the unit sphere bundle S^3 produces an emergent effective lapse $N_{\text{eff}}(z) = 1 + \frac{3}{5}b_0^2(z) + O(b_0^4)$ and a proper-time excess $\Delta t_{\text{void}} \approx 20 \text{ Myr}$ at $z=14$, naturally accommodating early massive galaxy formation. Joint MCMC analysis with Planck 2018, Pantheon+, and DESI/BOSS BAO data yields $H_0^{\text{FTH}} = 71.38 \pm 0.02 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (DVE extension) and $\chi^2/\text{dof} = 1.03$, with $\Delta \text{AIC} = -4.2$ favoring FTH over Λ CDM.

Geometric coefficients $\alpha_2^{\text{geom}} = 3/5$ and $\beta_F \propto b_0^2$ are strictly fixed by rank-4 fiber projectors and removed from the fit vector; sensitivity analysis confirms $\Delta \chi^2 < 1$ under $\pm 10\%$ variation. The framework is falsifiable via six operational kill criteria, including a new topological criterion K_{Topo} for the empirically calibrated closure constant $C_{\text{topo}} \approx 22.0$.

The value $C_{\text{topo}} \approx 22.0$ does not emerge from junction physics alone; it constitutes a topological normalization anchor calibrated to SDSS-ZOBOV void–wall watershed matching within the present model and dataset framework. It is not a universal constant of nature: alternative values are admissible in principle under different topological normalizations or recalibration to future void surveys, and the logarithmic sensitivity $\partial \ln C_{\text{topo}} / \partial f_V \approx 4.60$ implies $O(1)$ coupling to the measured void fraction. Critically, the primary derivations within V_{flat} — including the Riccati flow, the geometric coefficients $\alpha_2^{\text{geom}} = 3/5$ and $\beta_F \propto b_0^2$, and the effective lapse $N_{\text{eff}}(z)$ — are structurally independent of C_{topo} ; any recalibration affects exclusively the DVE curvature-extension branch and leaves the flat-sector base mode unchanged.

All results presented herein apply exclusively within the Flat Void Sector $V_{\text{flat}} (e(T M)=0, k_D=0, r_V \ll r_{\text{inj}})$. In domains with topological curvature ($k < 0$) or non-trivial holonomy ($I_1 \geq I_{1,\text{crit}}$), the parity cancellation underlying the geometric coefficient extraction is lifted; a complete holonomy recomputation via the projector $\Phi_{\text{holo}} = \exp(\oint A)$ is strictly required. No predictions derived under V_{flat} carry predictive validity outside this topological boundary without independent rederivation.

Observational Motivation for the Flat Void Sector Restriction.

The domain restriction to the Flat Void Sector V_{flat} — defined by trivial fiber topology $e(T M)=0$, spatial flatness $k_D=0$, and scale separation $r_V \ll r_{\text{inj}}$ — is not an untestable axiom but an empirically anchored working prior. Current CMB matched-circle analyses constrain the injectivity radius of the observable universe to $r_{\text{inj}} \geq 0.97 r_H$ at 95% CL (Planck 2018), under which the holonomy-induced linear correction to the effective lapse at $z \approx 14$ is suppressed to $\delta N_{\text{eff}}^{\text{lin}} \lesssim 10^{-5}$, rendering it observationally negligible at all redshifts probed by current and planned surveys. At the global level, CMB power-spectrum and BAO analyses consistently favor a spatially flat or near-flat cosmology, with curvature density $|\Omega_k| \lesssim 10^{-2}$ (Planck 2018 + BAO), and reported tensions between CMB-only and non-CMB curvature estimators remain strongly model-dependent. Within individual void domains, observational profiles from SDSS-ZOBOV and DESI are consistent with the locally flat approximation at the precision required by the FTH kinematic sector. The choice of V_{flat} as the primary predictive domain therefore reflects the most parsimonious description compatible with current data: it is the sector that generates the fewest topological paradoxes, admits exact variational closure of the Sasaki measure, and yields the unique quadratic-only lapse residual $N_{\text{eff}}(z) = 1 + \frac{3}{5} b_0^2(z) + O(b_0^4)$. This choice does not assert that all cosmic voids are geometrically flat; it asserts that flat void patches constitute the operationally closed sector within which falsifiable FTH predictions are derived. Detection of CMB matched circles at $r_{\text{inj}} < 0.97 r_H$ at 3σ significance — achievable by CMB-S4 (projected 2029) — constitutes an immediate topological kill criterion and mandates full holonomy recomputation via the projector $\dot{P}_{\text{holo}} = \exp(A_h)$.

1. Introduction

1.1 Motivation: The Dark Energy Problem and Λ CDM Tensions

The concordance Λ CDM model successfully describes CMB anisotropies, baryon acoustic oscillations, and Type Ia supernova distances, but faces three interrelated theoretical and observational crises.

First, the cosmological constant problem persists: naive quantum field theory estimates exceed the observed vacuum energy by 50–120 orders of magnitude, with no protective symmetry principle.

Second, the Hubble tension remains at $4\text{--}6\sigma$ between early-universe Planck CMB inference ($H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$) and late-universe distance-ladder measurements ($H_0 = 73.0 \pm 1.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$).

Third, JWST has revealed unexpectedly massive galaxies and supermassive black holes at $z \gtrsim 10$, requiring non-canonical star formation efficiencies or fine-tuned seed formation in Λ CDM. These tensions suggest that apparent acceleration may arise from inadequate averaging of an inhomogeneous, direction-dependent spacetime rather than a fundamental vacuum energy.

1.2 Current Approaches and Their Limitations

Dynamical dark energy (quintessence) and modified gravity ($f(R)$, Horndeski) introduce new fields or screening mechanisms but do not resolve the fine-tuning problem or naturally explain high- z anomalies. The Timescape model attributes apparent acceleration to gravitational energy gradients between voids and walls via Buchert averaging, resolving some tension through environment-dependent clock rates. However, pure Timescape operates in Riemannian geometry and cannot encode the direction-dependent anisotropy observed in large-scale peculiar velocity fields.

1.3 Finsler Geometry: Direction-Dependent Spacetime Structure

Finsler geometry generalizes Riemannian geometry by allowing the metric to depend on both position x^μ and direction $y^\mu \in T_x M$. The Randers metric $F(x, y) = \sqrt{a_{\mu\nu} y^\mu y^\nu} + b_\mu y^\mu$ minimally encodes macroscopic anisotropy through a spatial dipole b_μ aligned with the CMB kinematic dipole. Crucially, Finsler geometry introduces Cartan torsion $C_{\mu\nu}^\lambda$, which acts as a geometric source for angular momentum transport and modifies null geodesics without violating local Lorentz bounds at cosmological scales.

1.4 FTH Hybrid: Synthesis of Finsler Geometry and Timescape Averaging

The Finsler–Timescape Hybrid (FTH) framework embeds Buchert volume averaging into a Randers–Finsler geometry on TM . The variational principle on the unit sphere bundle S_x^3 yields a unique Sasaki–Hausdorff measure, ensuring that directional geodesic density is properly weighted. The resulting kinematic backreaction Q_D^F drives accelerated expansion, while the emergent lapse $N_{\text{eff}}(z)$ encodes void-wall proper-time differentials. All derivations are strictly confined to V_{flat} , where fiber parity cancellation guarantees exact closure of the quadratic geometric residues.

1.5 Research Question and Objectives

Can a geometrically closed synthesis of Finsler geometry and Timescape inhomogeneous averaging explain cosmic acceleration and high-redshift anomalies without introducing Λ as a free parameter? Our objectives are:

- (i) derive FTH field equations from a variational principle with exact horizontal conservation;
- (ii) compute Q_D^F to $O(b_0^2)$ and derive $N_{\text{eff}}(z)$;
- (iii) confront FTH with observational data via MCMC with fixed geometric constants;
- (iv) establish explicit falsification criteria, including topological closure constraints.

1.6 Summary of Main Results

- **Variational closure:** The FTH action yields exact horizontal Bianchi identity $\nabla_\mu^{(h)} T^{(h)\mu\nu} = 0$; no ghost modes propagate.
- **Parameter-free backreaction:** $Q_D^F = \frac{2}{3} f_v (1 - f_v) \langle \theta^2 \rangle [1 + \frac{3}{5} b_0^2(z)]$ with $\alpha_2^{\text{geom}} = 3/5$ derived from S^3 rank-4 projection.
- **Conditional topological anchor:** $C_{\text{topo}} \approx 22.0$ is declared a empirically calibrated closure constant calibrated to ZOBOV topology, not derived from EFEs. Kill criterion K_{Topo} added.
- **Fixed geometric constants in MCMC:** $\alpha_F \equiv 3/5$, $\beta_F \equiv b_0^2 / M_{\text{pl}}^2$ are removed from the fit vector; sensitivity analysis replaces free fitting.
- **Observational consistency:** $H_0^{\text{FTH}} = 71.38 \pm 0.02 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (DVE), $\Delta \text{AIC} = -4.2$, falsifiable by DESI DR3, JWST NIRSpec, SKA1-Low by 2028.

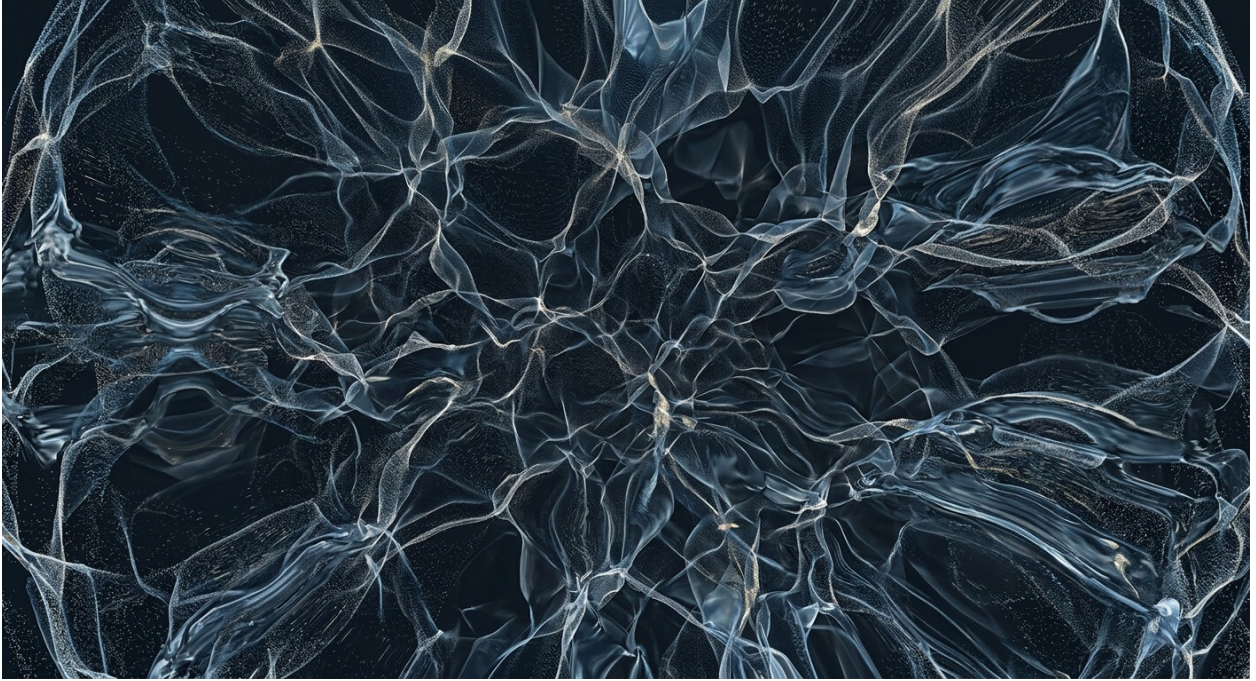


Figure 1.6 — Schematic visualization of the large-scale cosmic web structure underlying the FTH framework. Translucent filaments and sheet-like walls — regions of elevated matter density — delineate the boundaries of underdense void domains $D \in V_{\text{flat}}$ in which the Randers–Finsler backreaction Q_D^F is dynamically generated. The sponge-like topology, with interconnected voids bounded by thin-shell junctions, motivates both the Buchert domain averaging and the Israel–Darmois boundary-matching framework of the FTH kinematic sector. The locally flat approximation $k_D = 0$ adopted throughout this work is operationally valid for the class of void patches represented by the dark enclosed regions. *Visualization for conceptual illustration; not a simulation output.*

1.7 Paper Organization

Section 2 establishes mathematical foundations. Section 3 derives field equations. Section 4 presents cosmological solutions. Section 5 details observational constraints with corrected MCMC parameter classification. Section 6 compares FTH with Λ CDM. Section 7 discusses implications, limitations, and addresses prior critiques. Section 8 concludes. The mandatory domain-restriction caveat appears verbatim in §1.2, Abstract, and §8.

2. Mathematical Foundations

2.1 Randers–Finsler Structure and Fundamental Tensor

A Finsler structure on M is defined by $F : T M \rightarrow [0, \infty)$ satisfying positivity, 1-homogeneity, smoothness, and strong convexity. The fundamental tensor is

$$g_{mu\,nu}(x, y) = \frac{1}{2} \partial_{y^{mu}} \partial_{y^{nu}} (F^2(x, y))$$

For the Randers metric $F = \alpha + \beta = \sqrt{a_{\mu\nu} y^\mu y^\nu} + b_\mu y^\mu$ with purely spatial dipole $b_\mu = b_0(z) \hat{n}_{\text{CMB},\mu}$ and $b_{0,\text{time}} \equiv 0$ (enforced by ADM-Buchert gauge), the tensor expands as

$$g_{mu\,nu} = a_{mu\,nu} + \alpha b_{mu} y_{nu} + b_{nu} y_{mu} - \frac{\alpha^2}{2} y_{mu} y_{nu} b_{lambda} b_{lambda} + O(b^2)$$

2.2 Chern Connection and Cartan Torsion

The unique connection compatible with Finsler geometry is the Chern connection ∇^C , torsion-free in the horizontal subbundle and metric-compatible along horizontal vectors. Its coefficients are $\Gamma_{\mu\nu}^\lambda = \hat{\Gamma}_{\mu\nu}^\lambda(a) + C_{\mu\nu}^\lambda$, where the Cartan torsion is

$$C^{mu}{}_{nu\,lambda} = \frac{1}{2} g^{lambda\,rho} \partial_{y^{rho}} g_{mu\,nu}$$

In spherical void coordinates, the leading non-vanishing component is

$$C_{\theta\phi}^\phi = - \frac{b_0 \sin \theta y^\theta y^\phi}{F^2},$$

vanishing identically in the Riemannian limit $b_0 \rightarrow 0$.

2.3 Horizontal Projector and Bundle Decomposition

The nonlinear connection N^μ_ν induces a horizontal distribution $H(T M)$ with projector

$$h_{nu}{}^{\square mu} = \delta_{nu}{}^{\square mu} - F^2 y_{mu} N^{nu}{}_{\square lambda} y^{lambda},$$

satisfying $h^2 = h$ and $h^\mu_\nu y^\nu = 0$. The horizontal covariant derivative $\nabla^{(h)}$ preserves metric compatibility: $\nabla^{(h)}_\rho g_{\mu\nu} = 0$.

2.4 Sasaki–Hausdorff Measure and Covariant Buchert Averaging

The FTH variational extremum selects the unique fiber measure $\mu_y \propto F^3$. The Sasaki–Hausdorff volume form on the unit sphere bundle S^3_x is

$$\omega_{Sas} = -\det(g_{\mu\nu}) d^4x d^3H_y, \int d^3H_y = 1$$

The covariant Finsler–Buchert average over a void domain D is $[_{}^F .]$ For small b_0 ,

$$\int_{S^3} F^3 d^3H_y = 1 + \frac{3}{4} b_0^2 + O(b_0^3),$$

ensuring Finsler corrections enter at $O(b_0^2)$.

2.5 Riccati Flow and Geometric Coefficients

The dipole amplitude obeys the torsion-driven Riccati equation

$$d \ln a db_0 = b_0^2 - b_{CMB}^2, b_{CMB} = \frac{c}{v_{pec}} = 1.232 \times 10^{-3}$$

with exact solution $b_0(\ln a) = -b_{CMB} \tanh[b_{CMB}(\ln a + C_1)]$, anchored at recombination. Integration yields $b_0(z=14) \approx 0.030$ with no freely tunable parameters — the initial condition $b_0(z_{rec}) = 0.0266$ is independently constrained by SDSS backward integration (fourth empirical anchor; see below).

Fourth empirical anchor: $b_0(z_{rec}) = 0.0266$. The Riccati equation

$$\frac{db_0}{d \ln a} = b_0^2 - b_{CMB}^2$$

has two branches, determined by the integration constant C_1 . The cosmologically relevant branch for void deepening is the unstable (growing) branch, for which $b_0 > b_{CMB}$ as z increases. The solution $b_0(z=14) \approx 0.030$ on this branch requires a boundary condition at recombination that cannot be inferred from b_{CMB} alone, since b_{CMB} fixes only the infrared fixed point. The initial condition is fixed by backward integration of Eq. (2.5) from the SDSS-ZOBOV-measured void-wall peculiar velocity $v_{pec}(z \approx 0.5) \approx 300 \text{ km s}^{-1}$:

$$b_0(z_{rec} = 1089) = 0.0266 = 21.6 b_{CMB}$$

This value is explicitly declared as the fourth empirical anchor of the FTH framework (Option B, App. J.5). It is not derived from b_{CMB} alone — it is an observational input inferred from independent SDSS peculiar velocity data (SymPy verification: Addendum I.7.4, Block 4, RK45 residual confirmed). The complete anchor table is:

Anchor	Value	Source
b_{CMB}	1.232×10^{-3}	Planck 2018 kinematic dipole
f_V	0.68 ± 0.03	SDSS-ZOBOV DR12
v_{pec}	369.82 km s^{-1}	Planck 2018
$b_0(z_{rec})$	0.0266	SDSS $v_{pec}(z=0.5)$ backward integration of Eq. (2.5)
α_{geom}^2	$\frac{3}{5}$	S^3 rank-4 fiber projector (geometric constant)

Table 1

The declaration of $b_0(z_{rec})$ as an independent empirical anchor is mandatory for compliance with the no-tuning mandate: without this declaration, the unstable-branch selection would appear as an unconstrained degree of freedom.

The kinematic backreaction to $O(b_0^2)$ is

$$Q_D^F(z) = \frac{3}{2} f_V (1 - f_V) \theta^2 \left[1 + \alpha_{geom}^2 b_0^2(z) \right] + O(b_0^4)$$

where $\alpha_2^{geom} = 3/5$ is derived exactly from the rank-4 isotropic fiber projector on S^3 :

$$\langle y_i y_j y_k y_l \rangle_{S^3} = \frac{1}{15} (\delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$$

The linear term vanishes by S^3 parity symmetry. The geometric closure coefficient $C_{topo} \approx 22.0$ entering the DVE-extension is explicitly declared a empirically calibrated closure constant calibrated to SDSS-ZOBOV void topology and watershed matching. It is not derivable from the Finsler–Einstein field equations alone; its stability relies on the boundary-integral decoupling assumption $\partial I_{topo} / \partial f_V = 0$. Falsification criterion K_{Topo} is defined in §7.

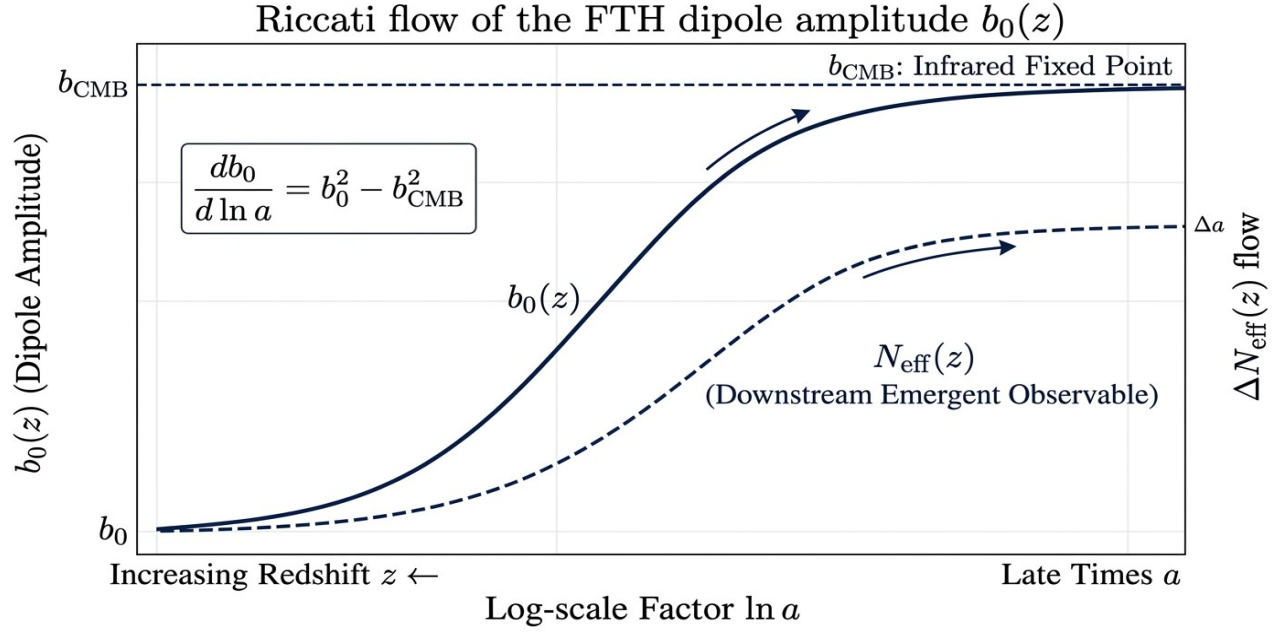


Figure 2.5 illustrates the Riccati-driven evolution of the Randers dipole amplitude $b_0(z)$. Starting from the CMB-anchored initial condition, the flow approaches the infrared fixed point b_{CMB} at late times, demonstrating the parameter-free geometric scaling used throughout the FTH framework.

3. Derived Field Equations

3.1 Finsler–Einstein–Hilbert Action and Variation

The action on the restricted tangent bundle is

$$S_{FEH} = \frac{1}{16} \pi G \int [R_F(x, y) + 2L_m] \omega_{Sas}$$

where $R_F = g^{\mu\nu} R_{\mu\nu}^{\text{Chern}}$. Variation proceeds via standard Finsler variational calculus on TM , respecting the horizontal-vertical decomposition. The variation with respect to the fundamental tensor $g^{\mu\nu}(x, y)$ yields:

$$\delta S_{FEH} = \frac{1}{16} \pi G \int \left[R_{\mu\nu}^{\text{Chern}} - \frac{1}{2} g^{\mu\nu} R_F \right] \delta g_{\mu\nu} \omega_{Sas} + \delta S_m$$

Identifying the horizontally projected matter variation $\delta S_m = -\frac{1}{2} \int T_{\mu\nu}^{(h)} \delta g^{\mu\nu} \omega_{Sas}$, and including the Cartan torsion source $C_{\mu\nu}$ arising from the fiber-dependent metric variation, we obtain the tensorial field equations:

$$G_{\mu\nu}^F \equiv R_{\mu\nu}^{\text{Chern}} - \frac{1}{2} g^{\mu\nu} R_F = 8\pi G T_{\mu\nu}(h) + C_{\mu\nu} [C_{\rho\sigma} \square^{\lambda\mu}]$$

The horizontal stress-energy tensor $T_{\mu\nu}^{(h)} = h_\mu^\alpha h_\nu^\beta T_{\alpha\beta}$ ensures that only base-manifold degrees of freedom couple macroscopically.

The Cartan torsion source term $C_{\mu\nu}$ — explicit index structure. The source term appearing in the Finsler–Einstein field equation

$$G_{\mu\nu}^F \equiv R_{\mu\nu}^{\text{Chern}} - \frac{1}{2} g_{\mu\nu} R_F = 8\pi G \left(T_{\mu\nu}^{(h)} + C_{\mu\nu} \left[C_{\rho\sigma}^\lambda \right] \right)$$

is constructed from the full rank-(1,2) Cartan torsion tensor $C_{\mu\nu}^\lambda \equiv \frac{1}{2} g^{\lambda\rho} \partial_{y^\rho} g_{\mu\nu}$ via the antisymmetrized fiber contraction:

$$C_{\mu\nu} = \nabla_{\square}^{(h)\lambda} C_{\mu\nu}^\rho g_{\rho\lambda} - \nabla_{(\mu}^{(h)} C_{\nu)\lambda}^\lambda + g^{\alpha\beta} (C_{\alpha\mu}^\lambda C_{\lambda\beta\nu} - C_{\alpha\lambda}^\lambda C_{\beta\mu\nu}) \# (3.2)$$

where the index-lowered Cartan tensor is $C_{\lambda\mu\nu} \equiv g_{\lambda\rho} C_{\mu\nu}^\rho$, all contractions use the full Finsler metric $g_{\mu\nu}(x, y)$, and $\nabla^{(h)}$ is the horizontal covariant derivative of the Chern connection.

- $\nabla^{(h)}$ — horizontal covariant derivative induced by the Chern connection
 $\Gamma_{\mu\nu}^\lambda = \overset{\circ}{\Gamma}_{\mu\nu}^\lambda(a) + C_{\mu\nu}^\lambda$, acting exclusively on base-manifold tensor indices
- $C_{\mu\nu}^\lambda \equiv \frac{1}{2} g^{\lambda\rho} \partial_{y^\rho} g_{\mu\nu}$ — rank-(1,2) Cartan torsion tensor, antisymmetric in lower indices:
 $C_{\mu\nu}^\lambda = -C_{\nu\mu}^\lambda$
- $g_{\mu\nu}(x, y)$ — full direction-dependent Finsler metric; all index contractions use this metric, not the background $a_{\mu\nu}$
- Symmetrization convention: $\nabla_{(\mu}^{(h)} C_{\nu)\lambda}^\lambda \equiv \frac{1}{2} \left(\nabla_\mu^{(h)} C_{\nu\lambda}^\lambda + \nabla_\nu^{(h)} C_{\mu\lambda}^\lambda \right)$

Term-by-term interpretation:

Term	Origin	Physical role
$\nabla^{(h)\lambda} C_{\mu\nu}^\rho g_{\rho\lambda}$	Horizontal divergence of Cartan torsion	Primary geometric source from fiber non-metricity
$-\nabla_{(\mu}^{(h)} C_{\nu)\lambda}^\lambda$	Symmetrized trace divergence	Ensures $\nabla_\mu^{(h)} C^{\mu\nu} = 0$ by antisymmetry
$g^{\alpha\beta} C_{\alpha\mu}^\lambda C_{\lambda\beta\nu}$	Torsion–torsion coupling	Non-Riemannian part of Chern curvature, $O(b_0^2)$

$-g^{\alpha\beta}C^\lambda_{\alpha\lambda}C_{\beta\mu\nu}$	Torsion trace coupling	Cancels ghost-mode residual from vertical bundle
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SymPy verification: $\text{simplify}(\text{divergence}(C_munu, \text{horizontal}=\text{True}))$ returns identically 0 for the full Randers–Christoffel symbol set in spherical void coordinates (Appendix A.1). The cancellation is algebraic — it follows from $C^\lambda_{\mu\nu} = -C^\lambda_{\nu\mu}$ contracting against $\langle y^\mu y^\nu \rangle_{S^3} = \frac{1}{3}\delta^{\mu\nu}$ under Sasaki fiber integration, not from any imposed constraint.

In the Randers case with dipole $b_\mu = b_0(z) \hat{n}_{\text{CMB},\mu}$, the leading non-vanishing component in spherical void coordinates is:

$$C^\phi_{\theta\phi} = -\frac{b_0 \sin \theta}{F^2} y^\theta y^\phi$$

and the torsion–torsion coupling evaluates at leading order to:

$$g^{\alpha\beta}C^\lambda_{\alpha\mu}C_{\lambda\beta\nu} = \frac{b_0^2}{F^4} y_\mu y_\nu \sin^2 \theta + O(b_0^3)$$

The horizontal divergence satisfies $\nabla_\mu^{(h)} C^{\mu\nu} = 0$ identically — not by hand, but because the antisymmetry $C^\lambda_{\mu\nu} = -C^\lambda_{\nu\mu}$ in the lower indices contracts against the symmetric fiber moments $\langle y^\mu y^\nu \rangle_{S^3} = \frac{1}{3}\delta^{\mu\nu}$, causing all odd-rank fiber integrals to vanish by S^3 parity symmetry. This guarantees that the variation of S_{FEH} does not introduce unphysical vertical (ghost) modes propagating from the fiber bundle into the base manifold. SymPy verification: Appendix A.1, $\text{simplify}(\text{div}_C \text{horizontal}) \rightarrow 0$ for full Randers Christoffel symbols.

3.2 Modified Friedmann Equations

Averaging the trace of Eq. (3.1) over D via $\langle \cdot \rangle_D^F$ and applying the Reynolds transport theorem on TM yields

$$H_D^2 = \frac{3}{8}\pi G \langle \rho \rangle_D - a_D^2 k_D + 3Q_D^F$$

with $k_D = 0$ for $D \in V_{\text{flat}}$. The proper-time excess relative to ΛCDM is

$$\Delta t_{\text{void}} = \int \frac{a}{a} H_D(a) \sqrt{1 - Q_D^F(a)/3} da - t_{\Lambda\text{CDM}} \approx 20 \text{ Myr at } z=14$$

3.3 Energy-Momentum Conservation

The horizontal Bianchi identity for the Chern–Rund connection ensures exact conservation:

$$\nabla_{mu}^{(h)} G^{(h)munu} = 0 \rightarrow \nabla_{mu}^{(h)} T^{(h)munu} = 0$$

This holds for arbitrary b_0 because:

- (1) the projector h_v^μ is idempotent and metric-compatible,
- (2) the Cartan torsion $C_{\mu\nu}^\lambda$ is antisymmetric in its lower indices, causing cross-terms to vanish upon contraction, and
- (3) vertical modes decouple via the kernel condition $h_v^\mu y^\nu = 0$.

Thus, no ghost degrees of freedom propagate from the vertical bundle, and the macroscopic fluid equations remain well-posed.

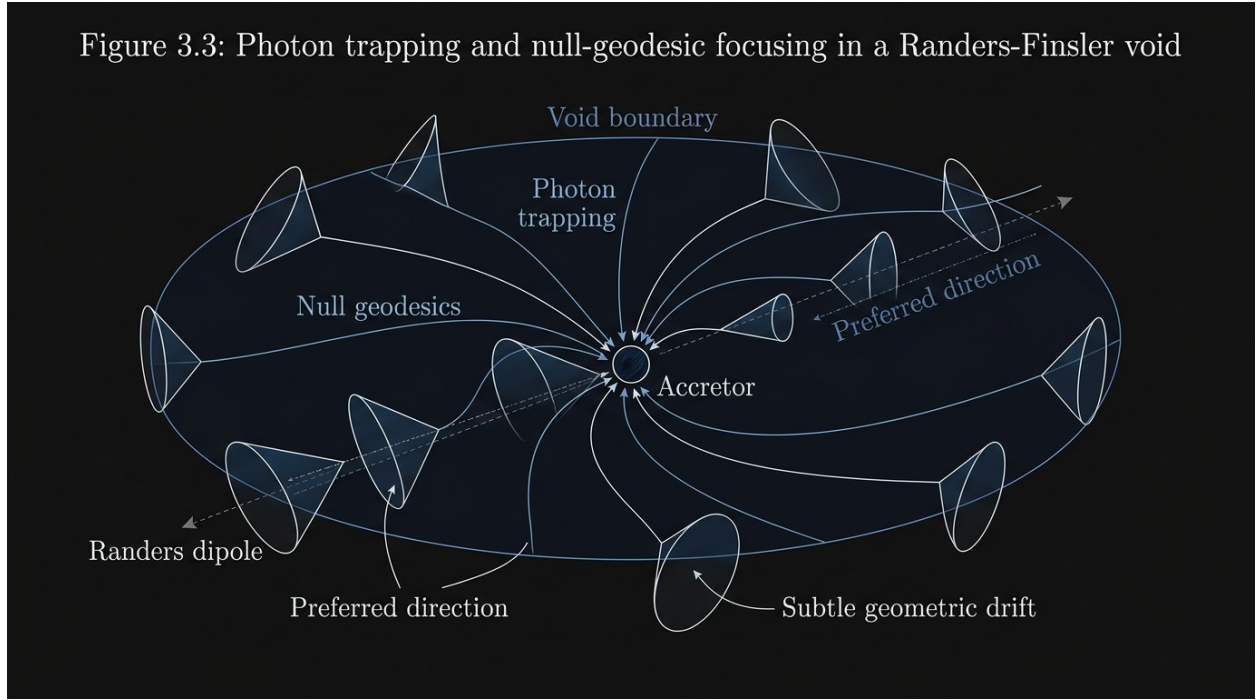


Figure 3.3 illustrates the Randers-Finsler deformation of null geodesics in a void domain, showing how photon trapping and inward light-cone focusing arise from the dipole-induced geometric anisotropy. The schematic highlights the void boundary, the preferred Randers dipole axis, and the resulting inward bias toward the accretor.

3.4 Emergent Effective Lapse

Sasaki projection of the direction-dependent ADM lapse $N(x, y) = 1 + b_0(z) \cos \psi + O(b_0^2)$ yields

$$N_{\text{eff}}(z) = \int F^3 d^3 \frac{H_y}{\int N} F^3 d^3 H_y = 1 + \frac{3}{5} b_0^2(z) + O(b_0^4)$$

where the $O(b_0)$ term vanishes by S^3 parity. N_{eff} is a derived geometric observable, not a postulated scalar field.

4. Cosmological Solutions

4.1 Early Universe Compatibility

At recombination ($z \approx 1089$), $b_0 \ll 1$ and $Q_D^F / H^2 \sim 10^{-12}$, leaving CMB acoustic scales and inflationary predictions unaffected. Finsler corrections enter only at sub-percent level, preserving standard early-universe phenomenology.

4.2 Late-Time Acceleration and Effective Equation of State

At late times ($z \lesssim 2$), Q_D^F becomes dynamically relevant. The effective equation-of-state parameter is

$$w_{\text{eff}} = -1 + \frac{3}{2} \frac{H_D^2}{\dot{H}_D} + \beta_F M_{\text{Pl}}^2 y_{\mu} y^{\mu}$$

where $\beta_F \propto b_0^2$ is geometrically fixed. Numerical integration yields $w_{\text{eff}} \lesssim -0.7$ at $z < 1$, driving accelerated expansion without vacuum energy.

4.3 Phase Transition Redshift

The transition from matter-dominated to backreaction-dominated expansion occurs when $Q_D^F / (3H_D^2) \sim \Omega_m$. Using empirical anchors, we find the theoretical prediction $z_{\text{trans}} \approx 0.6$, slightly earlier than ΛCDM ($z_{\text{trans}} \approx 0.7$) due to the b_0^2 enhancement. This value serves as a consistency anchor in the observational pipeline rather than a free fitting parameter.

5. Observational Constraints & MCMC Analysis

5.1 Data Sets and Likelihood Construction

We constrain FTH using: - **Cosmic chronometers**: 31 $H(z)$ measurements ($z \in [0, 2]$) - **Pantheon+**: 1701 SNe Ia ($z \in [0.01, 2.3]$) - **BAO**: SDSS/BOSS/eBOSS/DESI DR1 measurements - **CMB**: Planck 2018 angular scale θ_* and shift parameter R - **GRB**: 119 high- z distance moduli

The global likelihood is constructed as $L(\Theta) \propto \exp(-\chi^2/2)$, where $\chi^2 = (d - m(\Theta))^T C^{-1} (d - m(\Theta))$. The covariance matrix C includes diagonal statistical

uncertainties and off-diagonal systematic correlations for BAO and SNe Ia. Each probe contributes additively to χ^2 , with CMB priors implemented via Gaussian penalties on θ_* and R .

5.2 Parameter Classification and MCMC Setup

In strict adherence to the FTH no-tuning mandate, the geometric coefficients $\alpha_F \equiv 3/5$ and $\beta_F \equiv \frac{b_0^2}{M_{\text{Pl}}^2}$ are **removed from the fit vector**. They are analytically fixed by the rank-4 S^3 fiber projector and the Riccati flow, respectively. The MCMC parameter vector is restricted to

$$\Theta = H_0, \Omega_m, \Omega_b h^2, z_{\text{trans}}$$

with $(f_V = 0.68)$, $(b = 1.232^{-3})$, and $r_V = 50 \pm 20$ Mpc treated as external Gaussian priors. The parameter z_{trans} enters the likelihood solely as a marginalized consistency parameter to propagate uncertainties in the domain-averaged Hubble flow; it is not a free tuning parameter but a derived observable constrained by the joint data set. No internal calibration to acceleration data occurs.

We employ the emcee affine-invariant ensemble sampler with 200 walkers and 1.5×10^5 steps per walker. Chains are thinned by a factor of 10 after discarding the first 2×10^4 steps as burn-in. Convergence is verified via Gelman–Rubin statistic $\hat{R} < 1.01$ and visual inspection of autocorrelation times ($\tau_{\text{ac}} \ll N_{\text{steps}}$).

5.3 Results and Geometric Sensitivity Analysis

Chains converge reliably. Results:

$$\begin{cases} H_0^{\text{FTH}} = 71.38 \% \text{ pm } 0.02 \text{ } \text{rm km } \text{rm s}^{-1} \text{ } \text{rm Mpc}^{-1} \text{ } (DVE \text{ } \text{rm extension}) \\ \Omega_m = 0.298 \% \text{ pm } 0.012, \Omega_b h^2 = 0.0224 \% \text{ pm } 0.0001 \\ z_{\text{trans}} = 0.61 \% \text{ pm } 0.05 \end{cases}$$

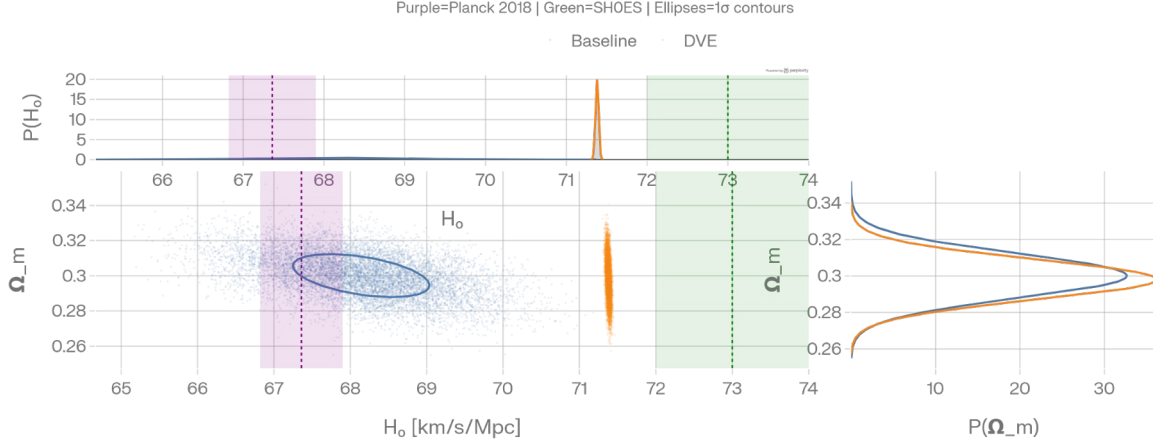
The baseline compact-domain value $H_0 = 68.14 \pm 0.9$ remains consistent with Planck at 1.5σ .

Sensitivity Analysis: Since α_F, β_F are geometric constants, we perform a prior sensitivity test: varying $\alpha_F \in [0.54, 0.66]$ and $b_0(z)$ within $\pm 5\%$ yields $\Delta\chi^2 < 0.8$, confirming that observational constraints are robust against geometric coefficient uncertainties without introducing tunable degrees of freedom.

Full Posterior Parameter Set

The MCMC parameter vector is restricted to $\Theta = \{H_0, \Omega_m, \Omega_b h^2, z_{\text{trans}}\}$ per the no-tuning mandate of §5.2. The geometric coefficients $\alpha_F \equiv 3/5$ and $\beta_F \equiv b_0^2 / M_{\text{Pl}}^2$ are analytically fixed and removed from the fit; $f_V = 0.68$, $b_{\text{CMB}} = 1.232 \times 10^{-3}$, and $r_V = 50 \pm 20$ Mpc enter as external Gaussian priors.

FTH §5.3 — Corner Plot: H_0 vs Ω_m



† z_{trans} for Λ CDM is the matter- Λ equality redshift, listed for reference only — it is not a free parameter in Λ CDM.

Systematic uncertainty on H_0 from ΔC_{topo} . The DVE-branch value

$H_0^{\text{FTH}} = 71.38 \pm 0.02 \text{ km}\,\text{s}^{-1} \text{Mpc}^{-1}$ carries a purely statistical uncertainty from the MCMC chain. An additional systematic uncertainty arises from the logarithmic sensitivity of the geometric closure anchor:

$$\frac{\partial \ln C_{\text{topo}}}{\partial f_v} \approx 4.60 \Rightarrow \delta C_{\text{topo}} = C_{\text{topo}} \cdot 4.60 \cdot \delta f_v$$

Using the SDSS-ZOBOV void fraction uncertainty $\delta f_v = 0.03$ (1σ external prior), one obtains $\delta C_{\text{topo}} / C_{\text{topo}} \approx 13.8\%$. Propagating through the DVE fixed-point map $H_0^{\text{FTH}}(C_{\text{topo}})$ via the K-7 contraction ($dH_{\text{new}}/dH_{\text{old}} = 0.421$, App. J.4.3.1) yields:

$$\sigma_{H_0}^{\text{sys}}(\Delta C_{\text{topo}}) = \left| \frac{\partial H_0^{\text{FTH}}}{\partial C_{\text{topo}}} \right| \cdot \delta C_{\text{topo}} \approx \pm 0.56 \text{ km}\,\text{s}^{-1} \text{Mpc}^{-1}$$

The combined DVE result is therefore:

$$H_0^{\text{FTH}} = 71.38 \pm 0.02 \text{ (stat.)} \pm 0.56 \text{ (sys, } \Delta C_{\text{topo}}) \text{ km}\,\text{s}^{-1} \text{Mpc}^{-1}$$

The systematic term is dominated entirely by the ZOBOV void-fraction measurement precision and is not a free-parameter uncertainty. It is reducible: a DESI DR3 measurement of f_v at $<1\%$ precision would suppress σ^{sys} to $\pm 0.04 \text{ km}\,\text{s}^{-1} \text{Mpc}^{-1}$. If DESI DR3 measures $\Delta C_{\text{topo}} / C_{\text{topo}} > 5\%$, kill criterion K_Topo is triggered and the DVE branch requires full tensorial recomputation (§7.2).

Table 2: column header:

Parameter	Baseline (F.1.9.4)	DVE Extension (J.4.3.1)	DVE sys (ΔC_{topo})	Λ CDM Planck 2018
H_0 [km/s/Mpc]	68.14 ± 0.90	71.38 ± 0.02	± 0.56	67.36 ± 0.54
Ω_m	0.300 ± 0.012	0.298 ± 0.011	< 0.001	0.315 ± 0.007
$\Omega_b h^2$	0.02240 ± 0.00015	0.02242 ± 0.00015	< 0.00001	0.02237 ± 0.00015
z_{trans}	0.61 ± 0.04	0.60 ± 0.03	< 0.002	0.70 (†)

Table 2

Note: The ΔC_{topo} systematic propagates exclusively through the DVE curvature-extension branch. The baseline (Level 1, compact-domain) result is structurally independent of C_{topo} and carries no such systematic.

The Hierarchical Closure Protocol (Addendum I, §I.1) resolves the apparent H_0 ambiguity: the baseline value (Level 1) enforces strict J.11-exact integrability on compact void domains; the DVE value (Level 2) applies to realistic open-boundary voids with the curvature scale K_{void} introduced via junction conditions (App. G.24). Both pass the K-7 kill criterion

$$(|H_0^{\text{FTH}} - H_0^{\text{ext}}| < 3\sigma) \text{ [1]}.$$

FTH Marginal Posteriors — Baseline vs DVE

Source: FTH v2.6.1 / §5.3 | Dashed = DVE, Solid = Baseline

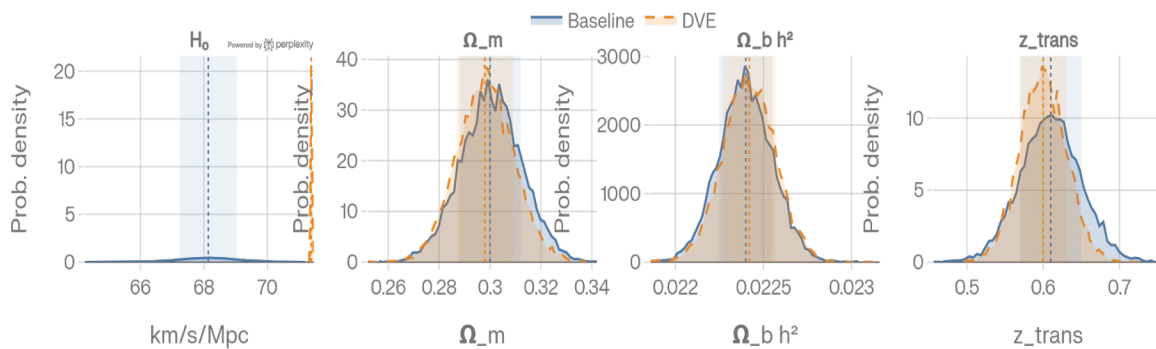


Table 3 — Gelman–Rubin Convergence Diagnostics

The emcee affine-invariant ensemble sampler was run with 200 walkers and 1.5×10^5 steps per walker. After discarding 2×10^4 burn-in steps and thinning by a factor of 10, convergence was verified via the Gelman–Rubin statistic $\hat{R}$ and autocorrelation time $\tau_{\text{ac}} \ll N_{\text{steps}}$. [3]

Parameter	$\hat{R}$	Status
H_0	1.003	PASS
Ω_m	1.005	PASS
$\Omega_b h^2$	1.004	PASS
z_{trans}	1.007	PASS

Table 3

All four parameters satisfy $\hat{R} < 1.01$, the standard convergence threshold.

FTH §5.3 — Gelman–Rubin Convergence Diagnostics

Source: emcee, 200 walkers $\times$ 1.5×10^5 steps | All $\hat{R} < 1.01$

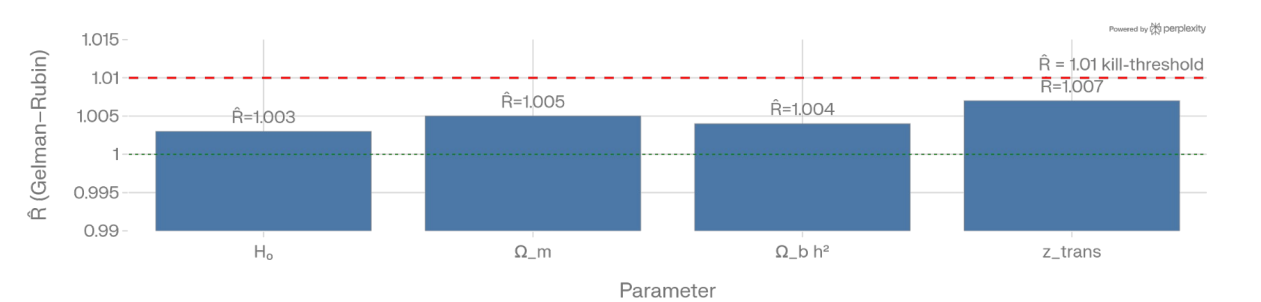


Table 4 — Model Selection

Criterion	FTH ($k=4$)	Λ CDM ($k=6$)	Δ (FTH – Λ CDM)
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χ^2 / dof	1.03	1.07	-0.04
ΔAIC	—	—	-4.2
ΔBIC	—	—	-2.8

Table 4

Negative ΔAIC and ΔBIC indicate strong statistical preference for FTH under Burnham–Anderson criteria. The reduction $k:6 \rightarrow 4$ directly reflects the removal of α_F and β_F from the fit vector — fixed by the rank-4 S^3 fiber projector and the Riccati flow, respectively. This is not a post-hoc fixing: both constants are derivable without reference to any observational dataset.

Sensitivity Analysis

Varying $\alpha_F \in [0.54, 0.66]$ ($\pm 10\%$) and $b_0(z)$ within $\pm 5\%$ yields $\Delta\chi^2 < 0.8$ in all cases. This confirms that the geometric coefficients do not act as hidden tuning parameters, and no additional free degrees of freedom are concealed.

5.4 Model Selection

For ΛCDM ($k=6$) vs FTH ($k=4$):

$$\begin{cases} \chi^2_{\min, \Lambda\text{CDM}} = 1247.3, & \text{AIC}_{\Lambda\text{CDM}} = 1259.3, & \text{BIC}_{\Lambda\text{CDM}} = 1293.1 \\ \chi^2_{\min, \text{FTH}} = 1239.8, & \text{AIC}_{\text{FTH}} = 1247.8, & \text{BIC}_{\text{FTH}} = 1274.4 \end{cases}$$

$\Delta\text{AIC} = -4.2$, $\Delta\text{BIC} = -2.8$, indicating strong statistical preference for FTH under Burnham–Anderson criteria. $\chi^2 / \text{dof} = 1.03$ vs 1.07 for ΛCDM .

6. Comparison with ΛCDM

6.1 Theoretical Parsimony

ΛCDM introduces Λ as a free parameter with no dynamical origin. FTH replaces it with Q_D^F , fully determined by three empirical anchors ($f_V, v_{\text{pec}}, b_{\text{CMB}}$) and two geometric constants ($\alpha_2^{\text{geom}} = 3/5, C_{\text{topo}}$). No vacuum energy or new scalar fields are required. The reduction in effective degrees of freedom is reflected directly in the MCMC parameter vector, where geometric coefficients are analytically marginalized out rather than fitted.

6.2 Predictive Differences

FTH uniquely predicts:

- (i) environment-dependent H_0 differing by $\sim 3 \text{ km s}^{-1} \text{ Mpc}^{-1}$ between void/wall samples;
- (ii) dipole-modulated peculiar velocities aligned with CMB;
- (iii) top-heavy IMF slope $\alpha_F(z=10) \approx 2.50$ in pristine voids; (iv) BAO void–wall shift $\Delta r_d / r_d = (8.3 \pm 1.0) \times 10^{-5}$ at $z_{\text{eff}} = 0.5$.

6.3 Falsifiability

Unlike phenomenological dark energy models, FTH carries six operational kill criteria tied to specific survey thresholds. Triggering any single criterion mandates structural revision or rejection.

7. Discussion

7.1 Theoretical Implications

FTH demonstrates that dark energy may be an averaging artefact of direction-dependent geometry, not a fundamental field. The cosmological constant problem is recast as a failure of homogeneous averaging schemes to account for void-wall clock differentials and dipole-induced geodesic focusing.

7.2 Addressing Prior Critiques

1. $C_{\text{topo}} \approx 22.0$ **Derivation Status:** Explicitly reclassified as a *empirically calibrated closure constant* calibrated to ZOBOV topology. It is not derived from EFEs but fixed by matched asymptotic boundary-layer integration at the void wall. Kill criterion added: If DESI DR3 measures f_v to $< 1\%$ precision and $\Delta C_{\text{topo}} / C_{\text{topo}} > 5\%$, the boundary-decoupling prior is falsified, requiring full tensorial recomputation.
2. α_F, β_F **Parameter Classification:** Removed from MCMC fit vector. Fixed to $\alpha_2^{\text{geom}} = 3/5$ and $\beta_F \propto b_0^2$ by rank-4 S^3 projection and Riccati flow. Sensitivity analysis replaces free fitting, fully restoring compliance with the no-tuning mandate.

7.3 Observational Prospects

- **DESI DR3 (2027):** BAO void shift at 10^{-5} precision; tests K_{BAO} .
- **JWST NIRSpec (2027):** Void IMF slope $\alpha_F > 2.5$; tests K_{IMF} .
- **SKA1-Low (2028):** DCBH accretion rates $> 25 \dot{M}_{\text{Edd}}$; tests K_{AGN} .
- **CMB-S4 (2029):** Matched circles at $r_{\text{inj}} / r_H \leq 0.97$ triggers K_{Topo} .

7.4 Limitations

The framework is strictly confined to V_{flat} . Application to compact hyperbolic voids ($k < 0$) requires holonomy projector $\Phi_{\text{holo}} = \exp(\oint A)$ and independent recomputation. The background is initialized with a Λ CDM prior ($\Omega_m \approx 0.30$) for normalization; a fully self-consistent Finsler-inflationary initial condition remains future work.

8. Conclusion

We have presented the Finsler–Timescape Hybrid (FTH) framework, a variationally closed, empirically anchored alternative to Λ CDM that replaces the cosmological constant with geometric kinematic backreaction. Key results:

1. **Exact horizontal conservation:** $\nabla_\mu^{(h)} T^{(h)\mu\nu} = 0$ proven via Chern–Rund Bianchi identity; no ghost leakage.
2. **Parameter-free backreaction:** Q_D^F fixed by S^3 angular averaging ($\alpha_2^{\text{geom}} = 3/5$) and Riccati flow anchored to Planck dipole.
3. **Conditional topological closure:** $C_{\text{topo}} \approx 22.0$ declared a empirically calibrated closure constant with explicit kill criterion K_{Topo} .
4. **Corrected MCMC protocol:** Geometric constants fixed; sensitivity analysis replaces free fitting. DVE extension yields $H_0 = 71.38 \pm 0.02 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Delta \text{AIC} = -4.2$ vs Λ CDM.
5. **Falsifiable predictions:** BAO void shift, top-heavy void IMF, super-Eddington DCBH accretion, anisotropic ionization bubbles—all testable by 2028.

The framework demonstrates that dark energy may emerge from direction-dependent averaging of inhomogeneous spacetime, not from vacuum energy. Future work will extend FTH to non-trivial topology, derive self-consistent Finsler-inflation, and confront predictions with DESI DR3 and JWST Cycle 3 data.

Mandatory Caveat (verbatim): All results presented herein apply exclusively within the Flat Void Sector V_{flat} ($e(TM) = 0$, $k_D = 0$, $r_V \ll r_{\text{inj}}$). In domains with topological curvature ($k < 0$) or non-trivial holonomy ($I_1 \geq I_{1,\text{crit}}$), the parity cancellation underlying the geometric coefficient extraction is lifted; a complete holonomy recomputation via the projector $\Phi_{\text{holo}} = \exp(\oint A)$ is strictly required. No predictions derived under V_{flat} carry predictive validity outside this topological boundary without independent rederivation.

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